

Stochastic Lanczos Quadrature for the AI-REML Score Traces

Pooled within-gene epistasis, matrix-free estimation (Simulation_code_MCREML_pooled_matfree)

1 The trace to be estimated

The AI-REML score for the two-component pooled model $V = \sigma_{\text{g} \times \text{g}}^2 W + \sigma_e^2 I$, with $K_1 = W$, $K_2 = I$ and $x = V^{-1}y$, is

$$s_i = \frac{1}{2}(x^T K_i x - \text{tr}(V^{-1} K_i)), \quad \mathcal{A}_{ij} = \frac{1}{2}(K_i x)^T V^{-1} (K_j x).$$

The data quadratics $x^T K_i x$ are exact once x is available, and the average information $\mathcal{A}$ is deliberately free of traces — that is the whole reason AI-REML is used instead of Fisher or observed-information scoring, both of which would need $\text{tr}(V^{-1} K_i V^{-1} K_j)$, a trace over a product of **two** inverses. So $\text{tr}(V^{-1} K_i)$ is the **only** stochastic object in the entire algorithm, and everything below concerns it alone.

The baseline: Hutchinson. With Nmc Rademacher probes $U = [u_1, \dots, u_{\text{Nmc}}] \in \{\pm 1\}^{n \times \text{Nmc}}$,

$$\text{tr}(V^{-1} K_i) \approx \frac{1}{\text{Nmc}} \sum_{l=1}^{\text{Nmc}} (V^{-1} u_l)^T (K_i u_l),$$

unbiased because $\mathbb{E}[uu^T] = I$. Each $V^{-1} u_l$ is a conjugate-gradient solve, and the solves must be **redone every REML iteration** because V moves with $(\sigma_{\text{g} \times \text{g}}^2, \sigma_e^2)$. At Nmc = 100 this is 100 of the 103 right-hand sides an iteration carries: essentially the whole cost of the estimator.

What SLQ changes. SLQ replaces the **evaluation** of each probe's quadratic form by a Gauss quadrature rule read off a Lanczos tridiagonalization. Applied naively to V this would still have to be redone every iteration and would buy nothing — a Lanczos run on V and a CG solve against V are the same iteration, and the Hutchinson estimate computed from a converged CG solve **already** is the Gauss rule (the Golub–Meurant identity $u^T V^{-1} u = u^T x_\ell$ for the ℓ -th CG iterate). The gain comes from running Lanczos on W instead, where the resulting rule turns out to be independent of the variance components and can therefore be built **once**.

2 The method

2.1 Gauss quadrature from Lanczos

Let $q_1 = u/\|u\|$ and let μ_q be the spectral measure of a symmetric matrix A at q_1 , so that for any smooth f

$$u^T f(A) u = \|u\|^2 \int f d\mu_q.$$

Running ℓ steps of Lanczos on A from q_1 produces an orthonormal basis Q_ℓ of the Krylov space $\mathcal{K}_\ell(A, q_1)$ and the symmetric tridiagonal $T_\ell = Q_\ell^T A Q_\ell$. Eigendecomposing $T_\ell = Y \Theta Y^T$ gives the ℓ -node **Gauss rule** for that measure:

$$\int f d\mu_q \approx \sum_{j=1}^{\ell} \tau_j f(\theta_j), \quad \theta_j = \Theta_{jj} \text{ (Ritz values)}, \quad \tau_j = (Y_{1j})^2 \text{ (squared first components)}.$$

The nodes and weights compress the whole spectral measure into ℓ numbers, and $\sum_j \tau_j = 1$ exactly.

2.2 Why Lanczos is run on W , not on V

This is the step that makes the method worth using here. V is **affine** in W ,

$$V = \sigma_{\text{gxc}}^2 W + \sigma_e^2 I,$$

so every power of V is a polynomial in W of the same degree and the two generate the **identical** Krylov space, $\mathcal{K}_\ell(V, q_1) = \mathcal{K}_\ell(W, q_1)$. Gram–Schmidt on that space returns the **same** orthonormal basis Q_ℓ , whence

$$T_\ell(V) = Q_\ell^T V Q_\ell = \sigma_{\text{gxc}}^2 Q_\ell^T W Q_\ell + \sigma_e^2 Q_\ell^T Q_\ell = \sigma_{\text{gxc}}^2 T_\ell(W) + \sigma_e^2 I.$$

A tridiagonal shifted and scaled this way has the **same eigenvectors**; only its eigenvalues move. Writing μ_j for the Ritz values of W ,

$$\theta_j = \sigma_{\text{gxc}}^2 \mu_j + \sigma_e^2, \quad \tau_j \text{ unchanged}.$$

So the quadrature obtained from W is not an approximation to the one obtained from V — it **is** that rule, re-parameterized. And since W does not depend on the variance components, (μ_j, τ_j) are computed **once** and serve every $(\sigma_{\text{gxc}}^2, \sigma_e^2)$ the optimizer ever visits.

2.3 Both traces from a single run

With $f(\theta) = 1/\theta$ for $K = I$ and $f(\theta) = \mu/\theta$ for $K = W$ (the extra factor is the eigenvalue of W at that node),

$$\begin{aligned} \text{tr}(V^{-1}I) &\approx \frac{1}{\text{Nmc}} \sum_l \|u_l\|^2 \sum_{j=1}^{\ell} \frac{\tau_j^{(l)}}{\sigma_{\text{gxc}}^2 \mu_j^{(l)} + \sigma_e^2}, \\ \text{tr}(V^{-1}W) &\approx \frac{1}{\text{Nmc}} \sum_l \|u_l\|^2 \sum_{j=1}^{\ell} \frac{\tau_j^{(l)} \mu_j^{(l)}}{\sigma_{\text{gxc}}^2 \mu_j^{(l)} + \sigma_e^2}. \end{aligned}$$

One Lanczos run yields **both** traces, so the fixed pre-loop product WU that the Hutchinson path needs is not required either.

A shortcut to avoid. Since $\sigma_{\text{gxc}}^2 W = V - \sigma_e^2 I$ exactly,

$$\sigma_{\text{gxc}}^2 \text{tr}(V^{-1}W) + \sigma_e^2 \text{tr}(V^{-1}I) = \text{tr}(I) = n,$$

so one could recover $\text{tr}(V^{-1}W) = (n - \sigma_e^2 \text{tr}(V^{-1}))/\sigma_{\text{gxc}}^2$ from the other trace. Algebraically correct, numerically bad: as $\sigma_{\text{gxc}}^2 \rightarrow 0$ both n and $\sigma_e^2 \text{tr}(V^{-1})$ tend to the same value and the expression becomes $0/0$. Carrying μ_j in the numerator evaluates the same identity stably, node by node, with no subtraction.

The identity is worth noting for a second reason: **both** estimators satisfy it **exactly** — Hutchinson because Rademacher probes have $u^T u = n$, SLQ because $\sum_j \tau_j = 1$. The two trace errors are therefore perfectly anti-correlated with ratio $-\sigma_e^2/\sigma_{\text{gxc}}^2$, and the identity is **not** available as an error diagnostic.

3 The steps

Setup (once per replicate, before the REML loop). Implemented in `slq_setup`:

1. Draw the Rademacher probes $U \in \{\pm 1\}^{n \times \text{Nmc}} - O(n \text{ Nmc})$. They are **fixed** for the whole run, so the objective stays a deterministic function of $(\sigma_{\text{gxc}}^2, \sigma_e^2)$.

2. For each probe, run ℓ steps of Lanczos on W from $q_1 = u_l / \|u_l\|$, accumulating the tridiagonal entries $\alpha_j = q_j^T W q_j$ and $\beta_j = \|\tilde{q}_{j+1}\|$. All Nmc probes share **one** W -apply per step — they differ only in their scalar recurrences — so the run costs ℓ applies of width Nmc, exactly the batching the CG solver already uses.
3. Reorthogonalize each new basis vector against all previous ones, twice. In floating point the Lanczos basis loses orthogonality as soon as a Ritz value converges, which duplicates nodes and corrupts the weights; full reorthogonalization at $O(n\ell^2)$ per probe is negligible beside the W -applies. A probe whose Krylov space is exhausted ($\beta_j \approx 0$) is frozen and its tridiagonal truncated at that point.
4. Eigendecompose each T_ℓ to get $(\mu_j, \tau_j) - O(\ell^3)$ per probe on an $\ell \times \ell$ tridiagonal, i.e. nothing. **Discard the Lanczos basis:** only the 2ℓ numbers per probe are kept.

Per REML iteration. Implemented in `slq_traces`: evaluate the two sums above — $O(\text{Nmc} \ell)$ arithmetic, **no linear solve and no matrix apply**. The iteration is then left with $1 + k = 3$ right-hand sides (one for $x = V^{-1}y$, $k = 2$ for $\mathcal{A}$) instead of $\text{Nmc} + 3$.

4 Complexity

Write c_W for the cost of applying W to one vector. For the pooled matrix-free kernel (see `Wu_complexity.typ`) that is $c_W = 2n \sum_{g=1}^G m_g^2$, which for G equal-size genes is $2nm^2/G$. Let t_{cg} be the CG iterations per solve, iters the REML iterations, ℓ the Lanczos steps, b the probe chunk size (bounds the basis memory).

Phase	Operation	Time	Space
setup	draw probes U	$O(n \text{ Nmc})$	$O(n \text{ Nmc})$
	ℓ Lanczos steps on W	$\ell \cdot \text{Nmc} \cdot c_W$	$O(n\ell b)$
	full reorthogonalization	$O(n\ell^2 \text{ Nmc})$	—
	tridiagonal eigenproblems	$O(\ell^3 \text{ Nmc})$	$O(\ell \text{ Nmc})$
per iter.	SLQ: both traces	$O(\text{Nmc} \ell)$	$O(1)$
	Hutchinson: $V^{-1}U$	$\text{Nmc} \cdot t_{\text{cg}} \cdot c_W$	$O(n \text{ Nmc})$

The Lanczos basis is the only bulky object and it is transient: probes are processed in chunks of b chosen so $n\ell b$ stays under a fixed buffer, and the basis is freed once (μ_j, τ_j) are extracted. Nothing of size $n \times n$ is ever formed, so the matrix-free character of the pipeline is preserved.

Total probe cost. Summing over the run, the two estimators spend

$$\text{Hutchinson : } \text{iters} \cdot t_{\text{cg}} \cdot \text{Nmc} \cdot c_W, \quad \text{SLQ : } \ell \cdot \text{Nmc} \cdot c_W,$$

a ratio of $\text{iters} \cdot t_{\text{cg}} / \ell$. At the pipeline's settings ($\text{iters} = 30$, $t_{\text{cg}} \approx 5$ with warm starts, $\ell = 25$) that is a factor ~ 6 ; the measured probe-column work fell from 5449 to 1149 columns, a factor 4.7. The whole estimation becomes

$$O\left(\underbrace{\ell \cdot \text{Nmc} \cdot c_W}_{\text{once}} + \underbrace{\text{iters} \cdot t_{\text{cg}} \cdot 3 \cdot c_W}_{\text{solves that remain}} + \underbrace{\text{iters} \cdot \text{Nmc} \cdot \ell}_{\text{traces, negligible}}\right),$$

against $O(\text{iters} \cdot t_{\text{cg}} \cdot (\text{Nmc} + 3) \cdot c_W)$ for Hutchinson.

5 Accuracy

The ℓ -node Gauss rule is exact for polynomials of degree $\leq 2\ell - 1$. For $f(\theta) = 1/\theta$ we have $f^{(2\ell)} > 0$ on $\theta > 0$, so the rule is a strict **lower bound** that increases monotonically with ℓ – raise ℓ until the traces stop moving.

Measured against Hutchinson **with exact solves** on the same probes (which isolates quadrature error from Monte-Carlo error), at $n = m = 1000$, $G = 10$:

ℓ	rel. error of $\text{tr}(V^{-1}W)$	rel. error of $\text{tr}(V^{-1}I)$
5	6.6×10^{-10}	1.6×10^{-10}
10	0	1.1×10^{-13}
25	1.1×10^{-13}	1.1×10^{-13}

Convergence is this fast because V is extremely well conditioned: W is positive semi-definite with $\text{tr}(W) = n$, and $\sigma_e^2 I$ dominates, so $\kappa(V)$ is a small constant at any realistic parameter value.

SLQ does not reduce the Monte-Carlo error. Its stochastic term is the **identical** Hutchinson variance $2 \sum_{i \neq j} A_{ij}^2 / Nmc$; ℓ controls only how exactly the same target is reproduced. Probe count, not quadrature, sets the statistical accuracy. (Methods that **do** cut probe count – Hutch++, XTrace – need a decaying spectrum to deflate, and here $V^{-1}W \approx I / (\sigma_{\text{gsg}}^2 + \sigma_e^2)$ is nearly flat, so they would gain essentially nothing. Rademacher Hutchinson is **exact** for a multiple of the identity, which is why the trace error is already only $\sim 0.05\%$.)

6 The conditioning guard

One cached rule must serve every variance setting the optimizer visits, and a fixed ℓ stops being enough once V becomes ill-conditioned – which happens as $\sigma_e^2 \rightarrow 0$, precisely the box-clamp boundary that a weakly-identified fit ($W \approx I$) is known to drift onto. Measured relative error of $\text{tr}(V^{-1}W)$ at $\ell = 25$:

$\kappa(V)$	rel. error	verdict
1.8	1.6×10^{-16}	safe
30.5	8.4×10^{-10}	safe
132	1.8×10^{-6}	safe
1.6×10^4	7.1×10^{-4}	comparable to the MC noise itself

The conditioning is available in **closed form** and costs nothing. Every interaction column of H is mean-centred, so $\mathbf{1}$ lies in the null space of W :

$$W\mathbf{1} = 0 \quad \Rightarrow \quad \lambda_{\min}(W) = 0 \quad \Rightarrow \quad \lambda_{\min}(V) = \sigma_e^2 \quad \text{exactly,}$$

verified numerically to 3×10^{-15} on both the dense and matrix-free paths. The upper end $\lambda_{\max}(V) \approx \sigma_{\text{gsg}}^2 \max_j \mu_j + \sigma_e^2$ is read off the cached Ritz values, which converge to the spectrum's extremes first. The classical Gauss/CG bound then predicts a relative error $\sim 2\rho^{2\ell}$ with $\rho = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, and `slq_reliable` falls back to exact probe solves for any iteration where that exceeds tolerance. The bound is conservative by about two orders of magnitude, which is what a guard should be: it errs toward falling back. On a run driven deliberately into the corner, the guard fired for 11 of 12 iterations and the result was bit-identical to the pure Hutchinson path.

7 Benefits

1. **The probe solves leave the REML iteration entirely.** The dominant per-iteration cost (Nmc of $Nmc + 3$ right-hand sides) becomes a one-off setup of ℓ applies. Per-iteration trace work drops to $O(Nmc \ell)$ scalar arithmetic.
2. **Both traces come from one run**, so the fixed product WU is no longer needed either.
3. **Exactly the same estimator.** SLQ changes how each probe's quadratic form is evaluated, not what is estimated or which probes are used. End-to-end over eight replicates the two paths agreed to 1.1×10^{-8} — the REML stopping tolerance, not an estimator difference.
4. **The deterministic objective is preserved.** Probes and quadrature nodes are both fixed, so the score remains a deterministic function of the parameters and the AI-Newton iteration converges to a fixed point rather than jittering.
5. **Robust by construction**, via the closed-form $\lambda_{\min}(V) = \sigma_e^2$ guard: accuracy is given up nowhere, speed is given up only where keeping it would cost accuracy.

The honest caveat. Measured wall-clock gain is $1.1 - 1.6 \times$, not the $4.7 \times$ the removed column count suggests. The reason is in `compute_WU_pooled`: it rebuilds the per-gene outer-product design D_g at $O\left(n \sum_g m_g^2\right)$ on **every call**, independent of block width — only the two gemms scale with the number of columns. Removing probe **columns** therefore does not remove proportional **time**: the call count fell only $156 \rightarrow 113$. The next real speed-up is to reduce the number of `compute_WU_pooled` **calls** (for instance by merging the per-iteration y and Kx solve groups into one batched call), not the number of probes.

Where the machinery pays for itself. Nothing in this pipeline currently computes a likelihood. The REML log-likelihood needs $\log \det V = \text{tr}(\log V)$, which **no** linear solve can produce at any tolerance and which the nodes computed here give immediately as $\sum_j \tau_j \log(\sigma_{\text{gxg}}^2 \mu_j + \sigma_e^2)$ — again for free at every parameter value, from the same cached rule. That would supply an actual convergence criterion (the current test, $\|\text{step}\| < \text{tol}$ under a Levenberg–Marquardt ridge and a box clamp, cannot distinguish a converged optimum from a damped step on a flat ridge), a likelihood-ratio test for $\sigma_{\text{gxg}}^2 = 0$, and a principled comparison between the pooled $1/P$ and per-gene $1/G$ kernel conventions.